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* Aim:
To simulate BCD to Excess-3 and Excess-3 to BCD code converters.

* Theory:

1. BCD codes:
Numeric codes represent numeric information i.e., only numbers as a series of 0's and 1's. Numeric codes used to represent decimal digits are called Binary coded Decimal (BCD) codes. A BCD code is one, in which the digits of a decimal number are encoded-one at a time into group of four binary digits. There are a large number of BCD codes in order to represents digits. So, 1,2,.... 9 it is necessary to use a sequence of at least four binary digits. Such a sequence of binary digits which represents a decimal digit is called code word.

2. Excess-3 code:
It is a non-weighted code. It is also a self-complementing BCD code used in decimal arithmetic units. The Excess-3 code for the decimal number is performed in the same manner as BCD except that decimal number 3 is added to

the each decimal unit before encoding it to binary.

* BCD to Excess-3

To convert from binary code A to binary code B, the input lines must supply the bit combination of elements as specified by code A and the o/p lines must generate the corresponding bit combination of code B. A combinational circuit performs this transformation by design 4-bit code, we must use four input variables and four output variables. Designate the four input binary variables by the symbols A, B, C, D and the four output variables by w, x, y and z. The truth table relating the input and output variables is as shown. A two-level logic diagram may be obtained directly from the Boolean expressions derived by the maps. The expressions obtained may be manipulated for the purpose of using common gates for two or more outputs. This manipulates illustrates flexibility obtained with multiple-output systems when implemented with three or more levels of gates.

BCD							
A	B	C	D	W	x	y	Z
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

→ Equations:

$Z = D$

$y = CD + C'D' = CD(C+D)'$

$x = B'C + B'D + BC'D'$

$\quad = B'(C+D) + BC'D'$

$\quad = B'(C+D) + B(C+D)'$

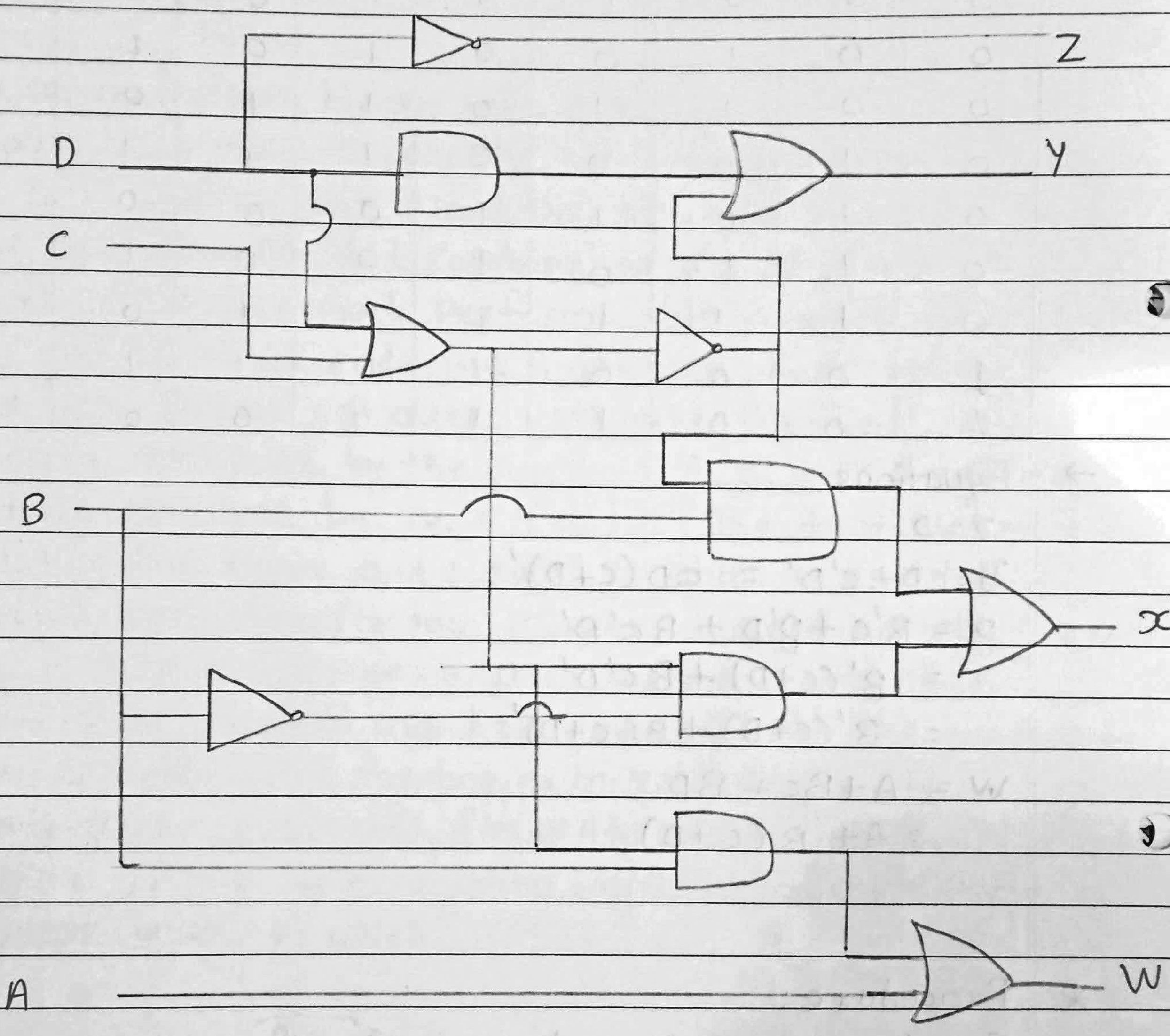
$W = A + BC + BD$

$\quad = A + B(C+D)$

* Procedure:

1. Select appropriate code converter from tab menu.
2. Run/execute the simulation by pressing the run button and observe the output of code converters on the output LED
3. Repeat the procedure for different inputs and note down the corresponding outputs.

→ Logic diagram for BCD to excess-3 code converter.



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Excess-3 to BCD

Excess-3 code can be converted back to BCD in the Same manner.
Let A, B, C and D be the bits representing the binary numbers where D is LSB and A is the MSB, and Let w, x, y and z be the bits representing the gray code of the binary numbers where z is the LSB and w is the MSB.
The truth table for the conversion is given below.

Decimal	Excess-3				BCD			
Digits	w	x	y	z	A	B	C	D
0	0	0	0	0	X	X	X	X
1	0	0	0	1	X	X	X	X
2	0	0	1	0	X	X	X	X
3	0	0	1	1	0	0	0	0
4	0	1	0	0	0	0	0	1
5	0	1	0	1	0	0	1	0
6	0	1	1	0	0	0	1	1
7	0	1	1	1	0	1	0	0
8	1	0	0	0	0	1	0	1
9	1	0	0	1	0	1	1	0

Equations:

$A = wx + wyz$

$B = x'y' + x'z' + xyz$

$C = y'z + yz'$

$D = z'$

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The corresponding digital circuit-
Here E_3, E_2, E_1 and E_0 correspond to w, x, y, z and B_3, B_2, B_1 and B_0. correspond to A, B, C and D.